Classical Mechanics

Lecture 4: Symmetry and Conservation Laws

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Chapter 1

Symmetry and Conservation Laws

Overview. Stationary action and the Euler–Lagrange equations are one half of the heart of classical mechanics; the Hamiltonian formulation is the other. Here we develop the first half far enough to expose a deep pattern: every continuous symmetry of a Lagrangian produces a conserved quantity. We shall discover the pattern in translations and rotations, prove it for a general infinitesimal transformation, and then examine carefully where the argument does and does not apply.

1.1 The Minimum Mathematics That Explains the Physics

The marks already on the board may look a little like the Schrödinger equation, but that is not tonight's subject. The subject is symmetry and conservation. There is no honest way to explain their connection without equations. The equations are elementary; what matters is arranging them so that the structure becomes visible. Scientific American may name the result, but here we want the minimum mathematics required to derive it as physicists would.

Introduce a small parameter ϵ. We retain terms proportional to ϵ and discard $O(\epsilon^2)$. This is shorthand for a limiting argument, not a claim that a nonzero real number literally has zero square. For an ordinary function $f(x, y)$,

$$\delta f = \frac{\partial f}{\partial x}\,\delta x + \frac{\partial f}{\partial y}\,\delta y + O(\epsilon^2), \tag{1.1}$$

where both δx and δy are proportional to ϵ. The symbol δf means the change in f, not the product of a number called δ with f. A stationary point obeys

$$\delta f = 0 \qquad \text{to first order.} \tag{1.2}$$

For a mechanical path $q_i(t)$, the analogous quantity is the action

$$S[q] = \int_{t_1}^{t_2} L(q, \dot{q}, t)\,dt. \tag{1.3}$$

Varying the path while fixing its endpoints gives

$$\delta S = \int_{t_1}^{t_2} dt\,\sum_i \left(\frac{\partial L}{\partial q_i}\,\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\,\delta \dot{q}_i\right). \tag{1.4}$$

The two terms arise because changing a path changes both its position and its velocity. Integration by parts, together with $\delta q_i(t_1) = \delta q_i(t_2) = 0$, turns $\delta S = 0$ into

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0. \tag{1.5}$$

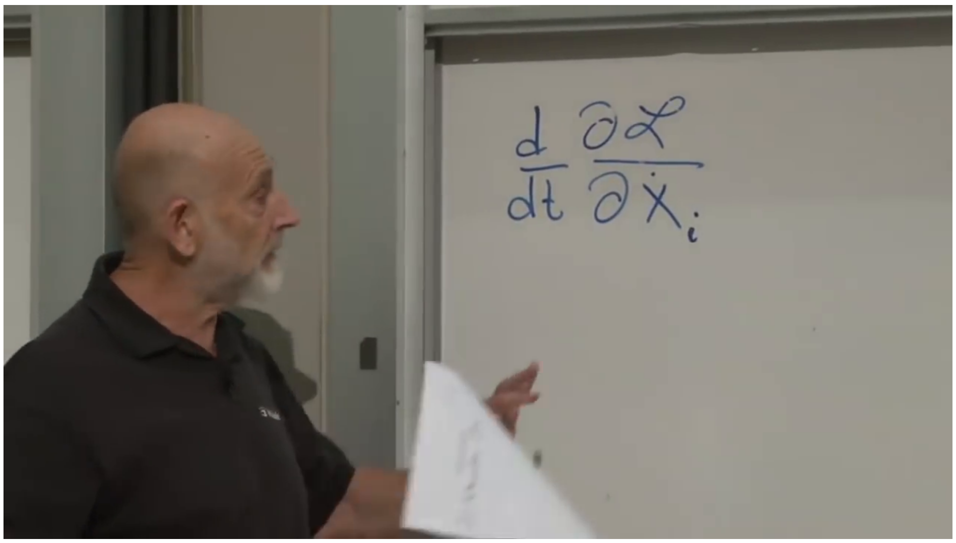

Figure 1.1: The Euler–Lagrange equation recalled as the starting point for symmetry and conservation.

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The correct phrase is *stationary action*, not necessarily least action. That distinction will return near the end.

1.2 Generalized Coordinates and Conjugate Momentum

Cartesian positions are too narrow a language for general mechanics. The coordinates may be particle positions, angles, collective variables, or field amplitudes, so we call them q_i. A Lagrangian is a compressed specification of the dynamics. Even the Standard Model, written out without very condensed notation, occupies roughly a page; writing a separate equation of motion for every field component would occupy vastly more. The one function L packages all of them.

The momentum conjugate to q_i is

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{1.6}$$

This is why generalized coordinates are traditionally called q's: the corresponding momenta are p's, giving us our p's and q's. In this notation Eq. (1.5) becomes

$$\dot{p}_i = \frac{\partial L}{\partial q_i}. \tag{1.7}$$

For the familiar one-dimensional particle,

$$L = \frac{m\dot{x}^2}{2} - V(x), \qquad \frac{\partial L}{\partial \dot{x}} = m\dot{x}, \tag{1.8}$$

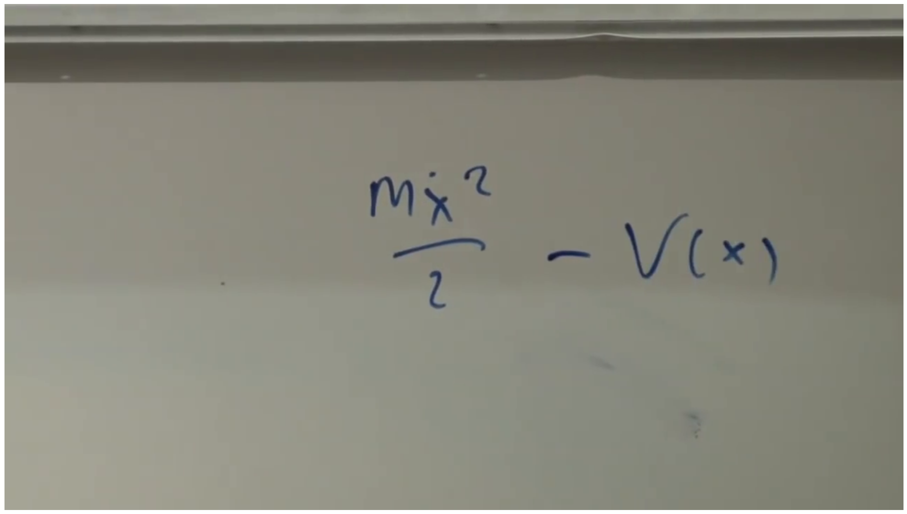

Figure 1.2: The kinetic-minus-potential Lagrangian used to identify conjugate momentum with $m\dot{x}$.
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so the conjugate momentum is ordinary momentum.

If q_i does not appear in L, then $\dot{p}_i = 0$. This is the familiar cyclic-coordinate route to a conservation law. The real connection is more general, and it is useful to discover it in examples before proving it.

1.3 Conservation Laws Found Before Their Explanation

1.3.1 Two coordinates and a separation potential

Set both masses equal to one and take

$$L = \frac{\dot{q}_1^2 + \dot{q}_2^2}{2} - V(q_1 - q_2). \tag{1.9}$$

One may picture two particles on a line. Their interaction depends only on their separation.

Writing V' for differentiation with respect to the argument of V,

$$\begin{aligned} \dot{p}_1 &= -V'(q_1 - q_2), \\ \dot{p}_2 &= +V'(q_1 - q_2). \end{aligned} \tag{1.10}$$

The second sign reverses because increasing q_2 decreases $q_1 - q_2$.

Question from the audience. Should the first momentum equation contain V', and where does its minus sign come from? [a]

Response. The Euler–Lagrange equation says $\dot{p}_1 = \partial L/\partial q_1$, not that the Lagrangian is zero. Since $L = T - V$, differentiating with respect to q_1 gives $-V'$. Differentiating with respect to q_2 supplies one more minus sign from $q_1 - q_2$, so the second equation has $+V'$.

[a] Lecture timestamp: 00:20:52–00:21:35.

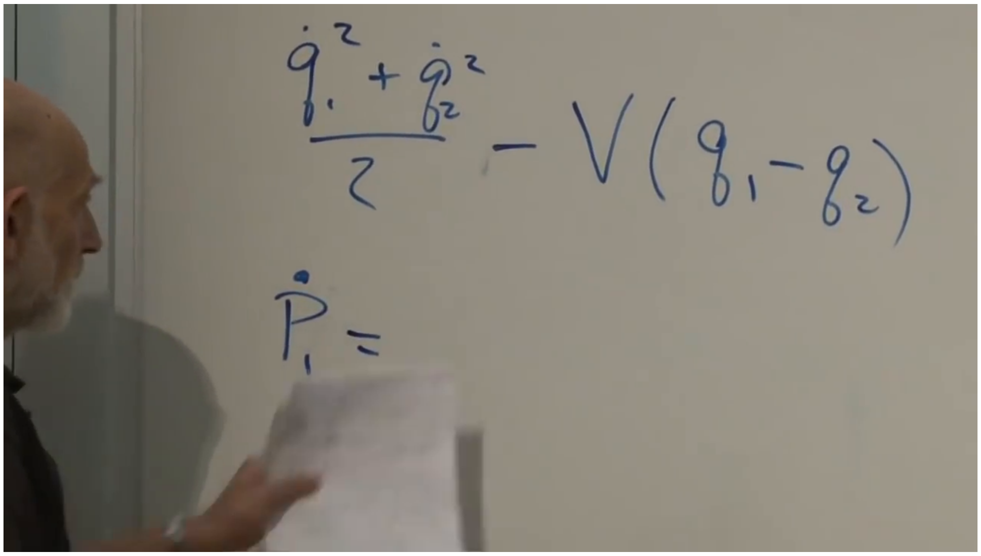

Figure 1.3: The two-coordinate Lagrangian and the beginning of its first momentum equation.

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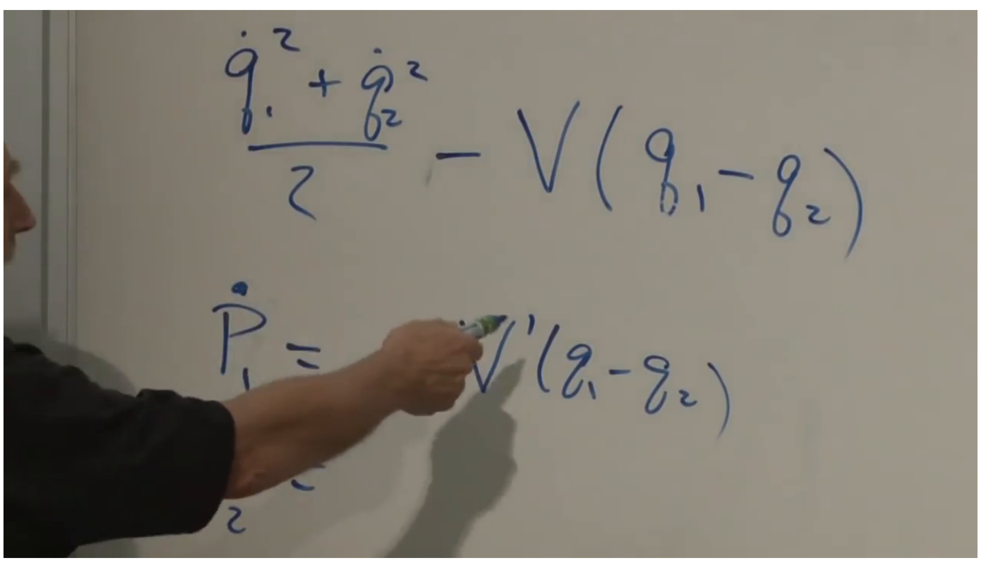

Figure 1.4: The sign-sensitive derivative in the p_1 equation; the p_2 equation receives the opposite sign.

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Adding Eqs. (1.10) gives

$$\frac{d}{dt}(p_1 + p_2) = 0. \tag{1.11}$$

We have found a conserved quantity, but have not yet explained why it exists.

1.3.2 A less obvious linear combination

Now let

$$L = \frac{\dot{q}_1^2 + \dot{q}_2^2}{2} - V(Aq_1 + Bq_2), \tag{1.12}$$

where A and B are constants. Then

$$\begin{aligned} \dot{p}_1 &= -A\,V'(Aq_1 + Bq_2), \\ \dot{p}_2 &= -B\,V'(Aq_1 + Bq_2). \end{aligned} \tag{1.13}$$

Multiplying the first equation by B, the second by A, and subtracting eliminates the force:

$$\frac{d}{dt}(Bp_1 - Ap_2) = 0. \tag{1.14}$$

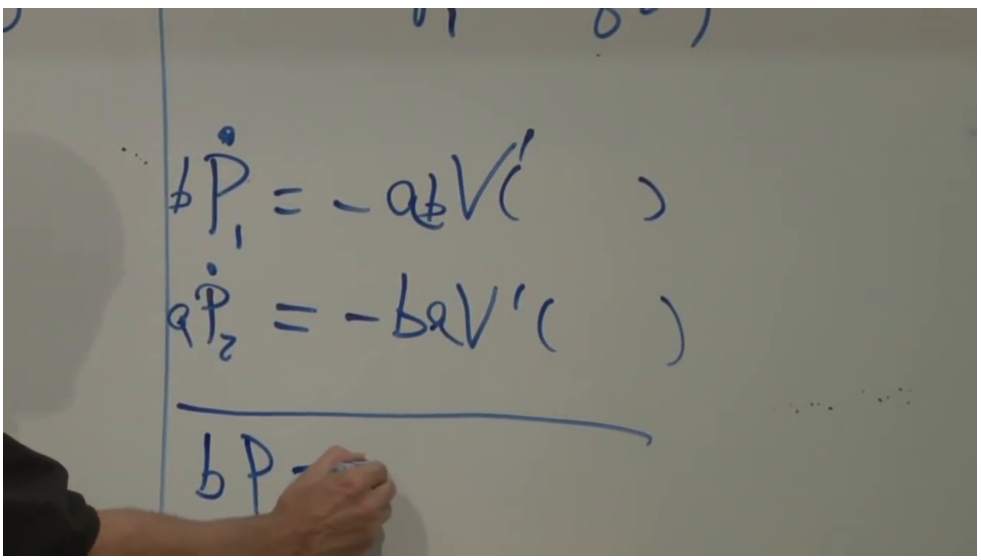

Figure 1.5: The weighted momentum equations whose force terms cancel in $Bp_1 - Ap_2$.
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Question from the audience. Where might constants such as A and B come from physically?[a]
Response. Consider two particles of masses m and M:

$$L = \frac{m\dot{X}^2}{2} + \frac{M\dot{Y}^2}{2} - V(X - Y).$$

Rescale to $Q_1 = \sqrt{m}\,X$ and $Q_2 = \sqrt{M}\,Y$. The kinetic term becomes $(\dot{Q}_1^2 + \dot{Q}_2^2)/2$, while the potential depends on $Q_1/\sqrt{m} - Q_2/\sqrt{M}$. Thus unequal masses naturally become coefficients in a linear combination. Replacing Q_2 by $-Q_2$ changes the displayed minus to a plus without changing its squared kinetic term.

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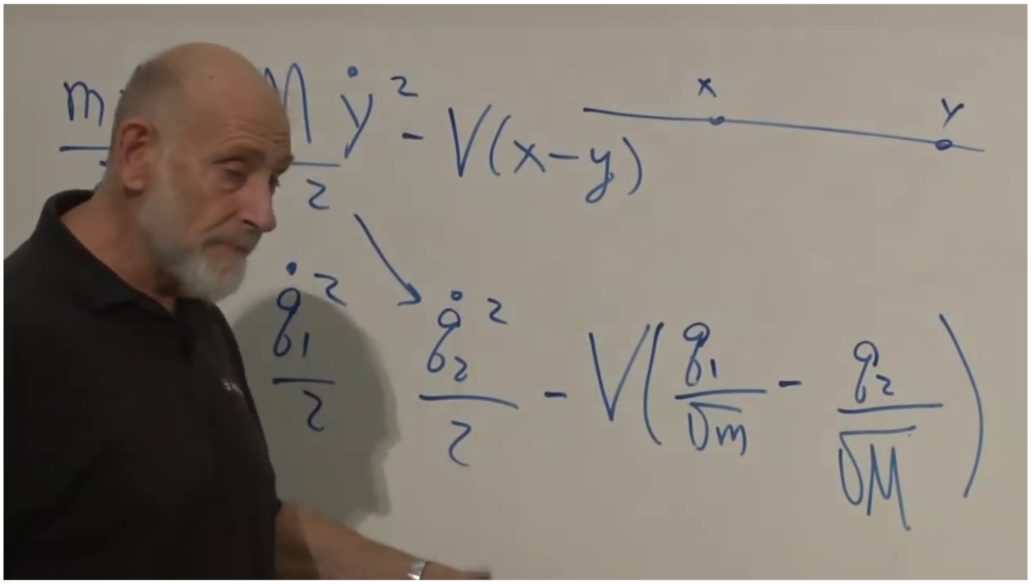

Figure 1.6: Mass-weighted coordinates simplify the kinetic energy while moving the mass factors into the potential.

Lecture frame: 00:28:40.

Question from the audience. Does this show that a conserved quantity can exist without a cyclic coordinate?[a]

Response. There is no *obvious* cyclic coordinate in the variables as written. A change of variables may reveal one, but hunting for it is not the useful viewpoint here. The symmetry argument will identify the conserved combination directly.

[a] Lecture timestamp: 00:31:13–00:31:58.

1.4 What We Mean by a Symmetry

1.4.1 Translations

Begin with a free particle of unit mass,

$$L = \frac{\dot{q}^2}{2}. \tag{1.15}$$

Translate every point of its history by the same constant:

$$q'(t) = q(t) + \epsilon. \tag{1.16}$$

Since ϵ is time independent, $\dot{q}' = \dot{q}$, and therefore $\delta L = 0$. This invariance is translation symmetry.

There are two equivalent pictures. Passively, shift the coordinate origin while leaving the object fixed. Actively, keep the axes fixed and move the object. The active language will prove less confusing: move the physical configuration and ask whether L notices.

For Eq. (1.9), translate both coordinates:

$$\delta q_1 = \epsilon, \qquad \delta q_2 = \epsilon. \tag{1.17}$$

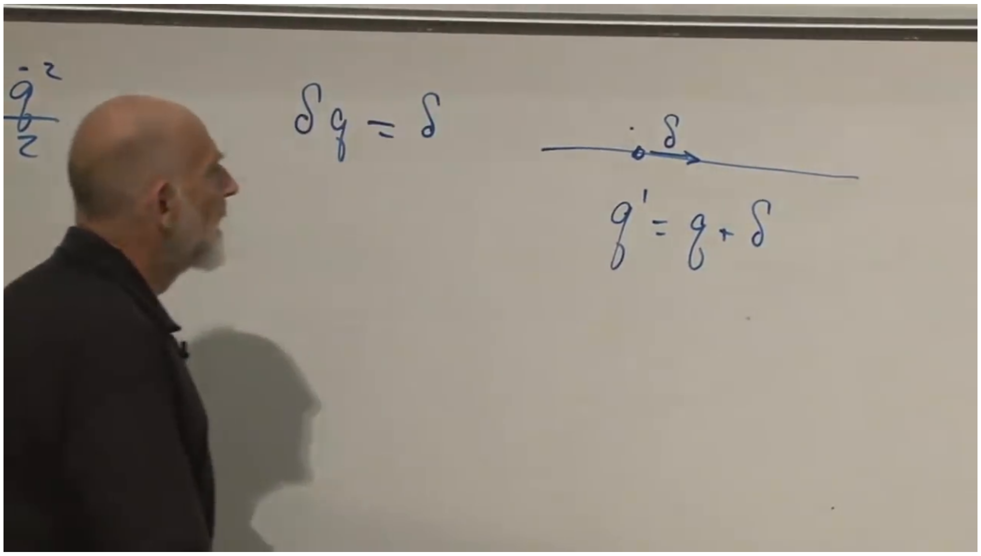

Figure 1.7: A common translation leaves the separation of two coordinates unchanged.
Lecture frame: 00:35:30.

The velocities are unchanged and so is $q_1 - q_2$.

For Eq. (1.12), use the less obvious transformation

$$\delta q_1 = B\epsilon, \qquad \delta q_2 = -A\epsilon. \tag{1.18}$$

Again the velocities do not change, while

$$\delta(Aq_1 + Bq_2) = AB\epsilon - BA\epsilon = 0. \tag{1.19}$$

Thus this transformation is a symmetry.

Question from the audience. Do the coefficients encode the masses, so that the conserved quantity becomes ordinary center-of-mass momentum? [a]

Response. Yes. In the mass-weighted variables the symmetry and conserved combination look asymmetric. Transforming back to X and Y recovers the ordinary simultaneous translation and total momentum.

[a] Lecture timestamp: 00:42:25–00:42:45.

1.4.2 Rotations

Consider one particle in the plane,

$$L = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) - V(x^2 + y^2). \tag{1.20}$$

This is a central potential. We compare two fixed coordinate systems rotated relative to one another; the axes are not rotating in time. With one consistent orientation,

$$\begin{aligned} x' &= x\cos\theta + y\sin\theta, \\ y' &= -x\sin\theta + y\cos\theta. \end{aligned} \tag{1.21}$$

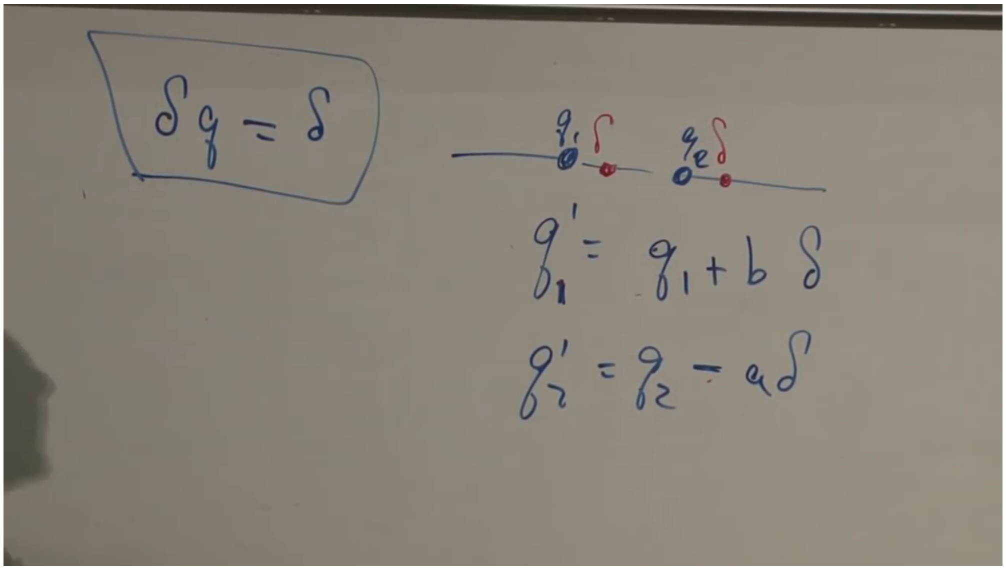

Figure 1.8: The asymmetric translation that preserves $Aq_1 + Bq_2$.
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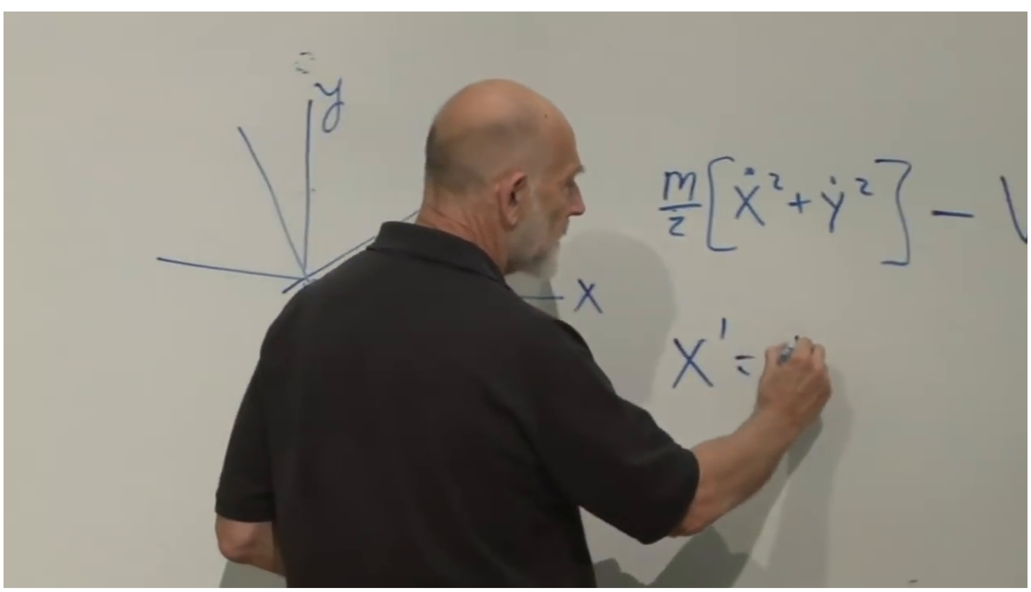

Figure 1.9: A central-potential Lagrangian and two fixed Cartesian frames related by a rotation.
Lecture frame: 00:45:40.

Set $\theta = \epsilon$ and keep first order:

$$\cos\epsilon = 1 + O(\epsilon^2), \qquad \sin\epsilon = \epsilon + O(\epsilon^3). \tag{1.22}$$

The infinitesimal rotation is

$$\begin{aligned} \delta x &= y\epsilon, & \delta y &= -x\epsilon, \\ \delta \dot{x} &= \dot{y}\epsilon, & \delta \dot{y} &= -\dot{x}\epsilon. \end{aligned} \tag{1.23}$$

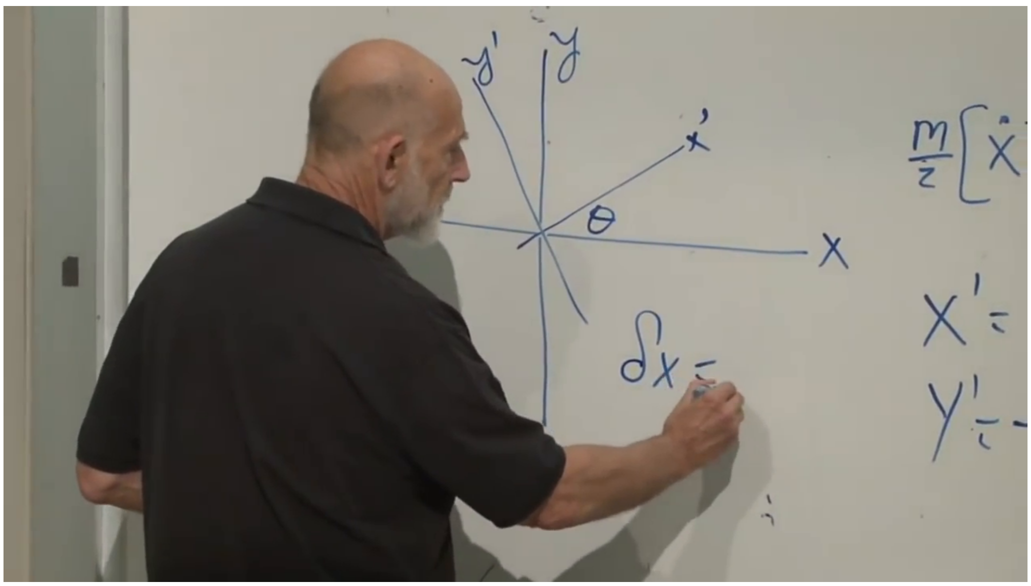

Figure 1.10: The infinitesimal rotation of coordinates and velocities.

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Unlike a translation, the amount by which x changes depends on y, and the amount by which y changes depends on x. Nevertheless,

$$\begin{aligned} \delta(x^2 + y^2) &= 2x\,\delta x + 2y\,\delta y = 0, \\ \delta(\dot{x}^2 + \dot{y}^2) &= 2\dot{x}\,\delta\dot{x} + 2\dot{y}\,\delta\dot{y} = 0. \end{aligned} \tag{1.24}$$

Both the potential and kinetic terms are invariant.

1.5 The General Conservation Theorem

The translation and rotation examples have one form in common. Let a continuous infinitesimal transformation be

$$\delta q_i = \epsilon f_i(q), \qquad \delta \dot{q}_i = \frac{d}{dt}\delta q_i, \tag{1.25}$$

where ϵ is time independent and f_i may depend on all the coordinates.

The first-order variation of the Lagrangian is

$$\delta L = \sum_i \left(\frac{\partial L}{\partial q_i}\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\delta \dot{q}_i \right). \tag{1.26}$$

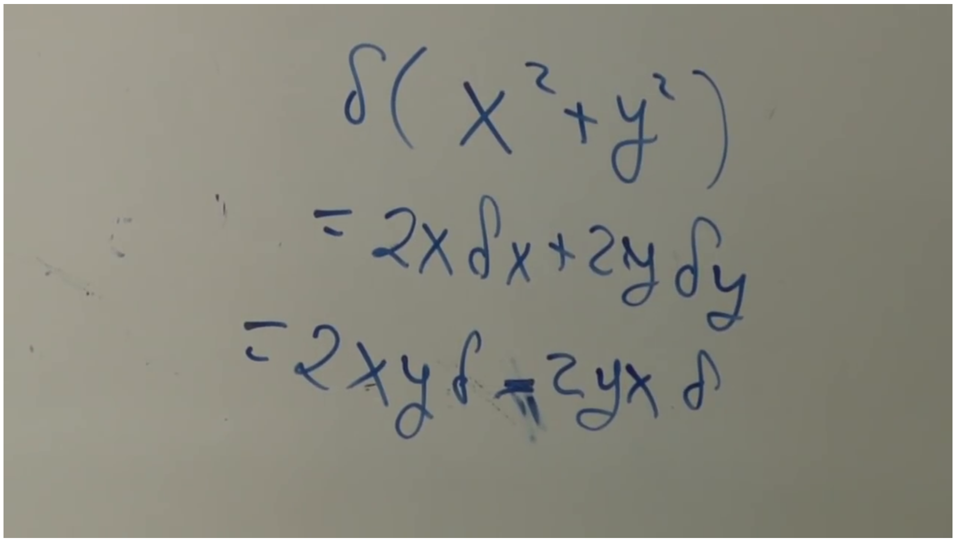

Figure 1.11: The first-order cancellation proving that the squared radius is rotationally invariant.
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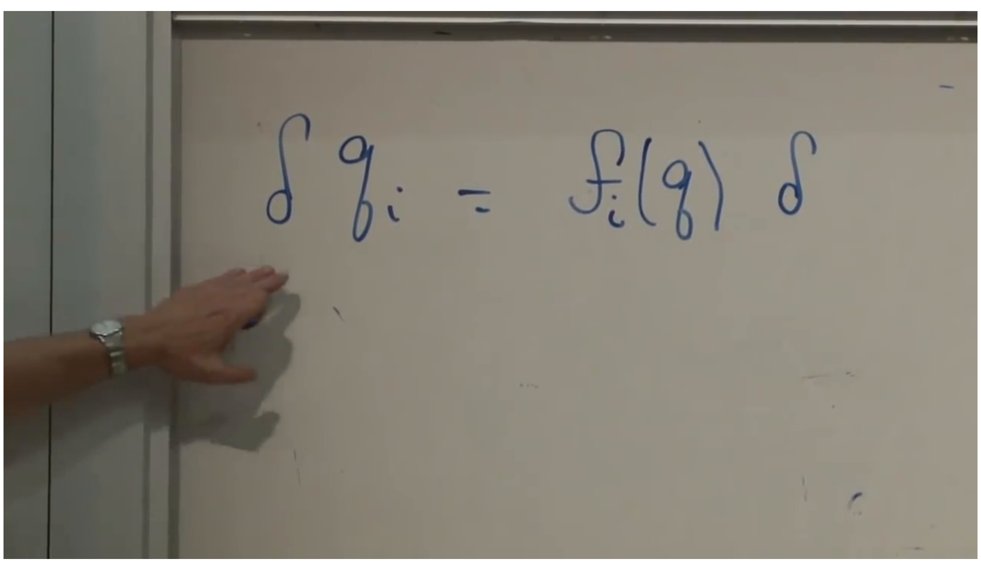

Figure 1.12: A general infinitesimal displacement field $\delta q_i = \epsilon f_i(q)$.
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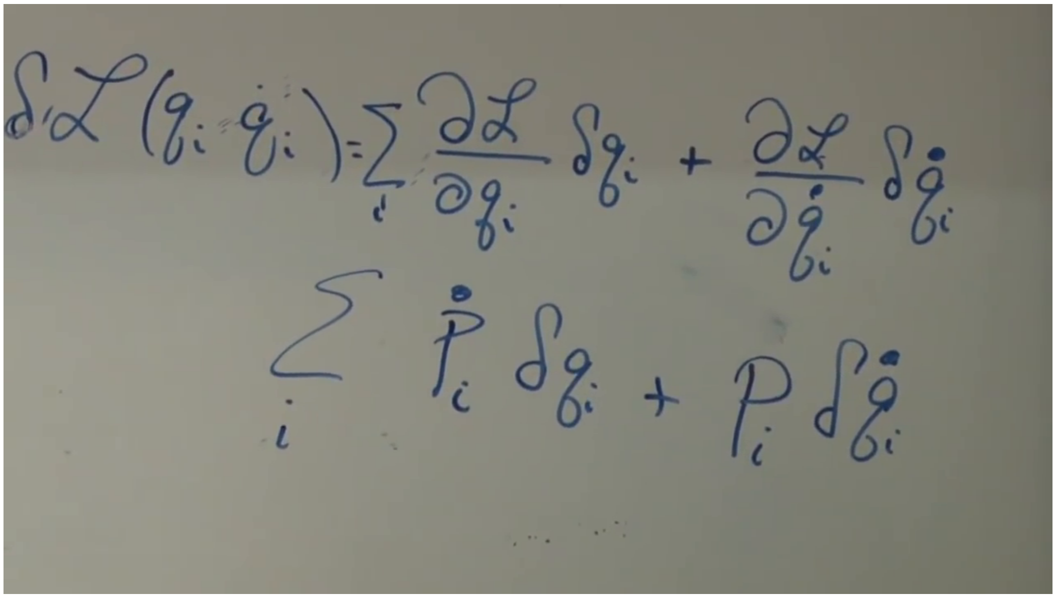

Figure 1.13: The two contributions to δL: coordinate variation and velocity variation.

Lecture frame: 01:02:30.

Now restrict attention to a physical path, so the Euler–Lagrange equations hold. Using

$$\frac{\partial L}{\partial \dot{q}_i} = p_i, \qquad \frac{\partial L}{\partial q_i} = \dot{p}_i, \tag{1.27}$$

we find

$$\begin{aligned} \delta L &= \sum_i (\dot{p}_i\, \delta q_i + p_i\, \delta \dot{q}_i) \\ &= \frac{d}{dt} \sum_i p_i \delta q_i = \epsilon \frac{d}{dt} \sum_i p_i f_i(q). \end{aligned} \tag{1.28}$$

The central step is only the product rule.

If the transformation is a symmetry, $\delta L = 0$. Since ϵ is an arbitrary constant factor,

$$\boxed{\frac{d\mathcal{Q}}{dt} = 0, \qquad \mathcal{Q} = \sum_i p_i f_i(q)}. \tag{1.29}$$

This is the classical-mechanics core of Noether's theorem for exact, time-independent point symmetries.

Question from the audience. In $f_i(q)$, does q carry another index? [a]

Response. The unindexed q denotes the whole set $(q_1, q_2, \ldots)$. Each component f_i may depend on all those coordinates. For rotation, for example, $f_x = y$ and $f_y = -x$.

[a] Lecture timestamp: 01:06:57–01:07:31.

The examples now cease to be accidents. A common translation has $f_1 = f_2 = 1$, so

$$\mathcal{Q} = p_1 + p_2. \tag{1.30}$$

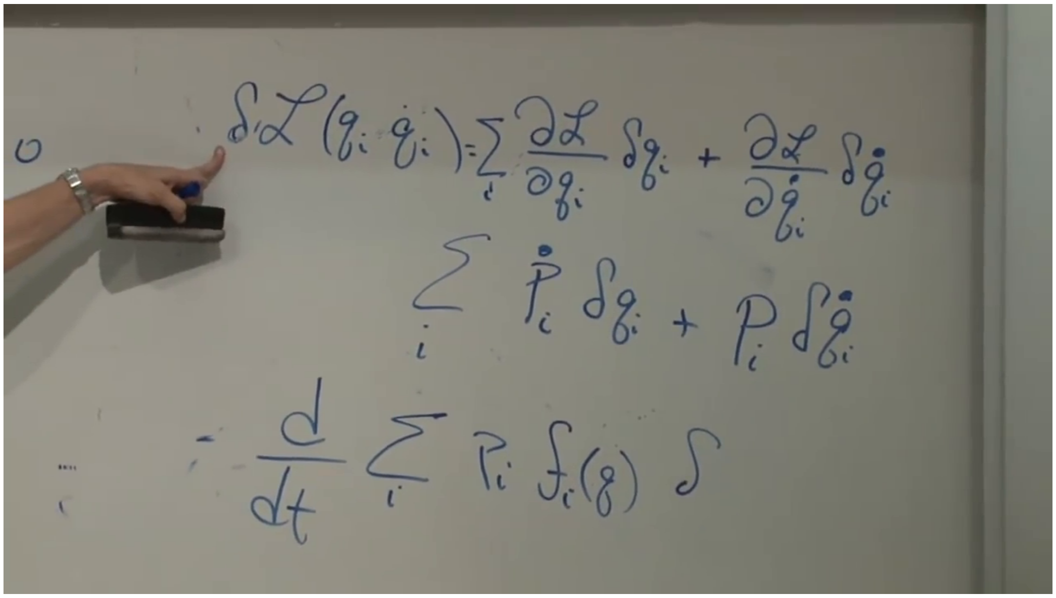

Figure 1.14: The product-rule step that turns invariance of L into a conserved quantity.
Lecture frame: 01:05:00.

The asymmetric translation has $f_1 = B$, $f_2 = -A$, so

$$\mathcal{Q} = Bp_1 - Ap_2. \tag{1.31}$$

For rotation, $f_x = y$ and $f_y = -x$:

$$\mathcal{Q} = p_x y - p_y x. \tag{1.32}$$

Up to the orientation sign, this is angular momentum,

$$J = xp_y - yp_x. \tag{1.33}$$

For many interacting particles, rotate every particle together. Invariance then conserves the sum of their angular momenta, not each particle's angular momentum separately.

1.6 Questions About the Theorem's Hypotheses

Question from the audience. Would a rotating set of axes introduce explicit time dependence into δL?[a]

Response. Yes. The proof just given assumes a time-independent symmetry parameter: a fixed translation or a fixed rotation. A transformation whose parameter depends on time produces additional terms and requires a more general treatment.

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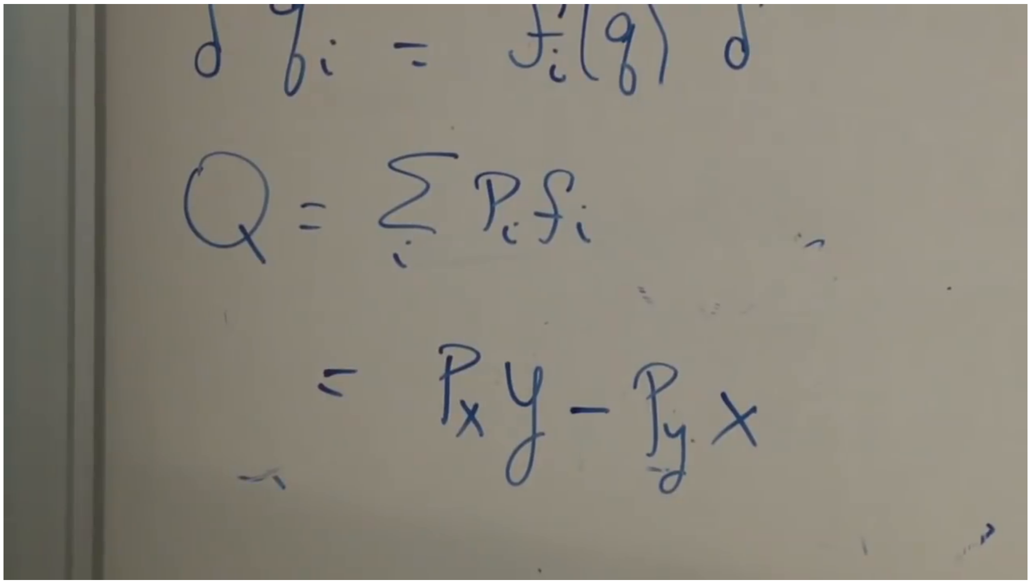

Figure 1.15: The rotational Noether charge, equal up to orientation to planar angular momentum.
Lecture frame: 01:11:00.

Question from the audience. Is the capital Q in the board formula the transformation itself?[a]

Response. No. That notation unfortunately collides with the coordinates q_i. The transformation is $\delta q_i = \epsilon f_i(q)$; the resulting conserved quantity is $\mathcal{Q} = \sum_i p_i f_i$. The coefficient f_i tells how the i-th coordinate moves under the symmetry.

[a]Lecture timestamp: 01:14:02–01:15:58.

Energy conservation is also generated by a symmetry, but a different one: translation in time, or independence of where the zero of time is placed. That case and the Hamiltonian formulation are deferred.

Question from the audience. Could the actual motion be absorbed into a changing coordinate system, so mechanics becomes nothing but coordinate transformations?[a]

Response. One can choose coordinates that follow a particular motion, but then the transformation depends on the initial state and on the trajectory. The symmetry transformation used here is fixed independently of which solution the system follows. Motion itself is more naturally represented in the combined space of coordinates and momenta.

[a]Lecture timestamp: 01:17:43–01:20:01.

1.7 The Harmonic Oscillator in Phase Space

With mass and spring constant set to one,

$$L = \frac{\dot{q}^2}{2} - \frac{q^2}{2}. \tag{1.34}$$

The conjugate momentum and equation of motion give

$$p = \dot{q}, \qquad \dot{p} = -q, \tag{1.35}$$

or

$$\dot{q} = p, \qquad \dot{p} = -q. \tag{1.36}$$

These equations describe uniform circular motion in the q-p plane. The back-and-forth real-space oscillator is the projection of that circle onto the q-axis: speed is largest at $q = 0$ and vanishes at the turning points.

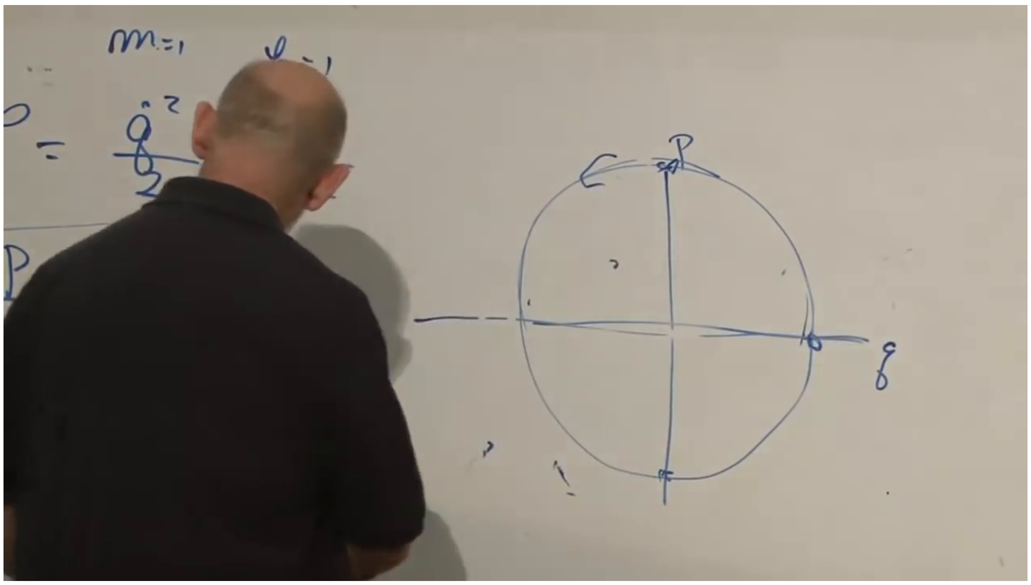

Figure 1.16: The harmonic-oscillator equations and their circular orbit in q-p space.
Lecture frame: 01:25:30.

Question from the audience. Does the phase-space circle run clockwise? [a]

Response. With p horizontal and q vertical, yes. At maximal positive q, $\dot{p} = -q < 0$, so the motion points downward. Reversing the placement of the axes reverses the visual sense, which explains why the convention is easy to remember incorrectly.

[a] Lecture timestamp: 01:25:47–01:27:15.

The pair in Eq. (1.36) already resembles Hamilton's equations, but it has been obtained entirely within Lagrangian mechanics.

1.8 Continuous and Discrete Symmetries

Question from the audience. Must a symmetry be expressed by an infinitesimal Cartesian transformation?[a]

Response. Cartesian coordinates are only convenient; the essential requirement for this proof is continuity. Some symmetries have no infinitesimal version. For an even potential, reflection sends $q \mapsto -q$ and $\dot{q} \mapsto -\dot{q}$, leaving L unchanged, but a reflection cannot be assembled from arbitrarily small reflections. This classical derivation therefore produces no associated conserved quantity.

[a] Lecture timestamp: 01:27:48–01:30:31.

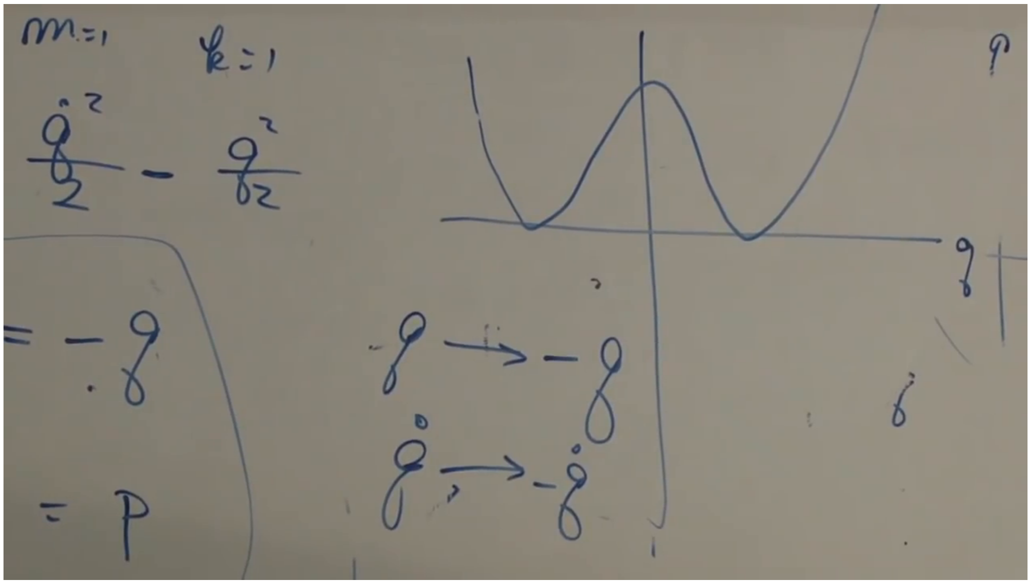

Figure 1.17: An even potential is invariant under the discrete reflection $q \mapsto -q$.
Lecture frame: 01:29:30.

Question from the audience. If the conserved quantity $\mathcal{Q}$ is nonzero, does that show the transformation was not a symmetry? [a]

Response. No. A conserved quantity may have any constant value. The relevant test is $\dot{\mathcal{Q}}$. If the quantity derived from a proposed transformation is not conserved, then that proposed transformation was not a symmetry under the hypotheses of the proof.

[a] Lecture timestamp: 01:30:40–01:31:30.

Question from the audience. Reflection still leaves the Lagrangian unchanged, so why is there no conservation law from it?[a]

Response. Invariance alone is visible, but the derivation passed through a first-order displacement and the product rule. Reflection moves one side of configuration space to the other in a single discrete step. With no infinitesimal generator f_i, the formula $\mathcal{Q} = \sum_i p_i f_i$ has nothing to use.

[a] Lecture timestamp: 01:31:34–01:32:27.

Question from the audience. What is the name for the property of being built from arbitrarily small transformations?[a]

Response. It is a *continuous symmetry*. Translation and rotation are continuous; reflection is a *discrete symmetry*.

[a] Lecture timestamp: 01:32:36–01:33:23.

Question from the audience. If many infinitesimal steps make a finite transformation, can the discarded $O(\epsilon^2)$ errors accumulate? [a]

Response. The infinitesimal invariance must hold at every point reached along the transformation, not only at the starting point. Divide a finite parameter interval into N steps of size ϵ: the accumulated first-order displacement is $N\epsilon$, while a local $O(\epsilon^2)$ remainder accumulates as $N\epsilon^2 = (N\epsilon)\epsilon \to 0$. This is the local-to-finite structure formalized by continuous transformation groups.

[a]Lecture timestamp: 01:33:23–01:36:10.

Question from the audience. Are discrete symmetries physically realized in classical systems?[a]

Response. Certainly. The harmonic oscillator itself has the even potential $V(q) \propto q^2$, so it is invariant under $q \mapsto -q$. There are many such examples; the point is only that this discrete invariance supplies no new classical Noether charge.

[a]Lecture timestamp: 01:36:32–01:37:25.

Question from the audience. But if both continuous and discrete transformations leave L unchanged, is that not the same condition? [a]

Response. It is the same finite statement of invariance, but not the same mathematical input to this proof. The proof needs a first-order variation. For a reflection, 'first order' has no meaning, so the chain of equations leading to a charge cannot even be started.

[a]Lecture timestamp: 01:37:37–01:39:18.

Question from the audience. How does this relate historically to Noether's theorem? [a]

Response. This is a restricted part of Emmy Noether's much more general theorem. Particular links between symmetries and conserved quantities were known earlier, but Noether placed them in a unified and remarkably general framework in the early twentieth century. Her mathematical importance was not matched by fair academic treatment.

[a]Lecture timestamp: 01:39:26–01:40:42.

Question from the audience. Is classical mechanics still an active research subject? [a]

Response. Yes. The elementary theorem used here is old, but nonlinear dynamics, integrable systems, chaos, fluids, and geometric mechanics remain rich subjects. The symmetry principle also continues far beyond classical particle mechanics.

[a]Lecture timestamp: 01:40:47–01:42:37.

1.9 Why the Action Principle Matters

Question from the audience. Is there an intuitive reason that symmetry should imply conservation?[a]

Response. Not from symmetry alone. Consider viscous drag,

$$m\ddot{x} = -c\dot{x}.$$

The equation is invariant under $x \mapsto x + \epsilon$, yet with $p = m\dot{x}$,

$$\dot{p} = -\frac{c}{m}p, \qquad p(t) = p(0)e^{-ct/m},$$

so momentum decays. This is an effective equation for a subsystem, not a fundamental stationary-action law for x alone. Include the fluid molecules and the rest of the closed system, and total momentum is conserved. Symmetry and conservation are tied together through the Lagrangian action principle, not through the appearance of a differential equation by itself.

[a]Lecture timestamp: 01:42:49–01:47:57.

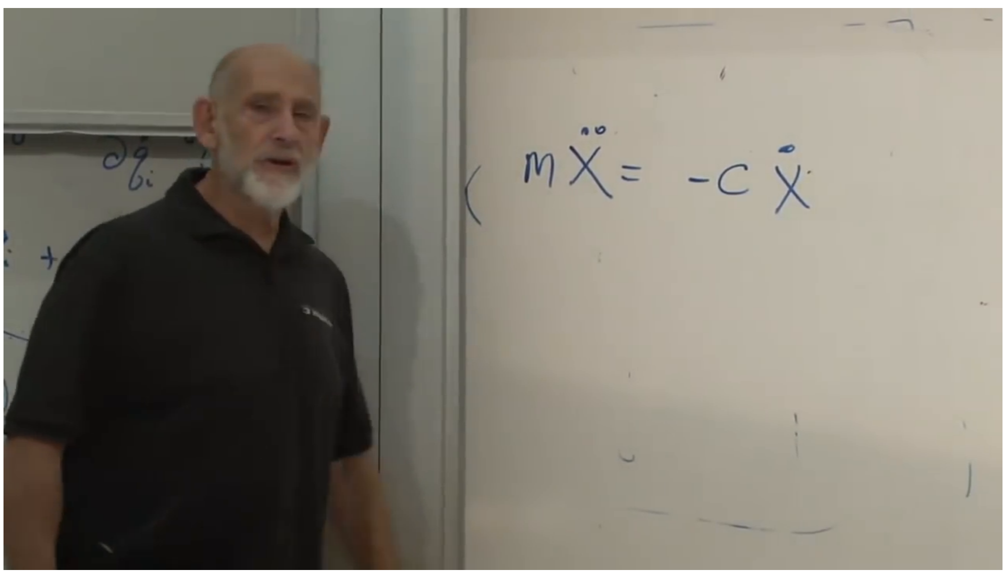

Figure 1.18: Translation-invariant viscous drag, whose reduced particle momentum nevertheless decays.

Lecture frame: 01:44:30.

Question from the audience. Can we quantify when the physical path is not a minimum of the action?[a]

Response. A true minimum is exceptional; the action is generally a saddle in path space. For the unit oscillator,

$$S[q] = \frac{1}{2}\int (\dot{q}^2 - q^2)\, dt.$$

A narrow, rapidly changing deformation raises the positive $\dot{q}^2$ contribution. A broad deformation that remains slightly displaced for a long time can make the negative $-q^2$ contribution dominate and lower the action. The same stationary path therefore has increasing directions and decreasing directions. Quantum mechanics, through the path-integral viewpoint, gives the deeper origin of stationarity, but that argument is deferred.

[a]Lecture timestamp: 01:48:08–01:51:38.

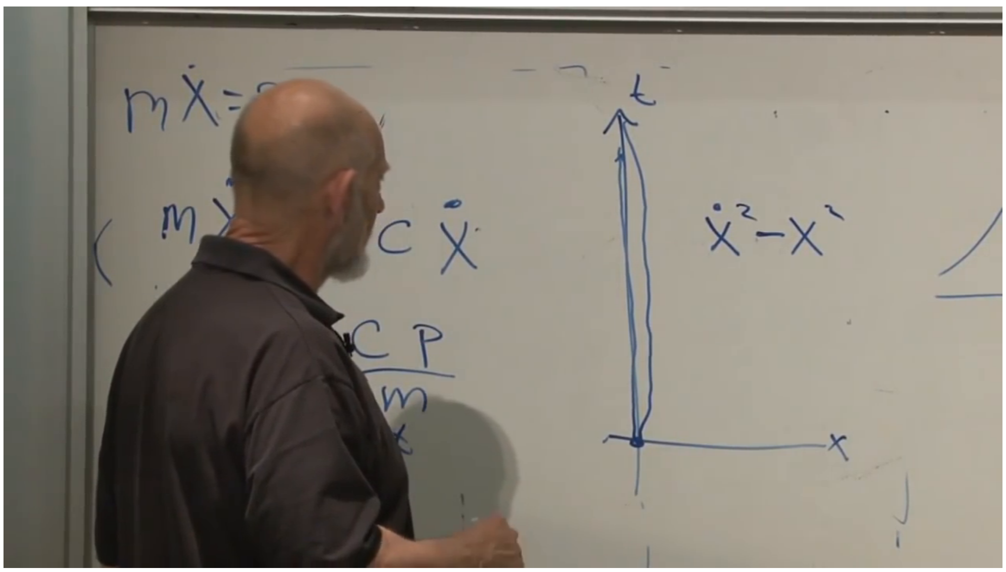

Figure 1.19: A broad neighboring oscillator path illustrates why stationary action is typically a saddle rather than a minimum.

Lecture frame: 01:50:20.

Question from the audience. Is there a one-to-one correspondence between continuous symmetries and conservation laws?[a]

Response. Within the variational setting and after identifying equivalent or trivial cases, that is essentially the right picture. Every continuous symmetry has a charge, and conserved quantities can generally be related back to symmetries. Precise formulations carry technical qualifications, so the broad statement should not be used carelessly.

[a]Lecture timestamp: 01:51:44–01:52:43.

Question from the audience. Must every Lagrangian possess a nontrivial symmetry? [a]

Response. No. The examples were chosen to have symmetries. A generic Lagrangian need not be invariant under any proposed spatial transformation.

[a]Lecture timestamp: 01:52:43–01:53:23.

Question from the audience. Should a symmetry finally be understood as an active motion or a passive change of coordinates? [a]

Response. Either description can be made equivalent, but the active one is cleaner here. Move the whole physical system and ask whether the Lagrangian changes. In passive language one must instead say that the *form* of the Lagrangian is unchanged when the coordinate frame moves.

[a] Lecture timestamp: 01:53:25–01:55:08.

Question from the audience. What comes next?[a]

Response. Conservation of energy, its origin in time-translation symmetry, and then the Hamiltonian formulation.

[a] Lecture timestamp: 01:55:10–01:55:35.